Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



## Team Details

a. Team name: Agamya
b. Team leader name: Siddhi Pandirkar
c. Problem Statement: Future of Work & Enterprise Productivity

# Brief about the idea

## When an expert leaves, the *why* leaves with them. It has happened for centuries.

### Bengal, muslin

One of history's finest cloths, spun from phuti karpas cotton. The cotton and the skills are gone.

*The "cut thumbs" story is legend; the record shows tariffs and machine-made imports. [1]*

### Sikkim, Majhi (माझी)

2013: India's last known speaker, Thak Bahadur Majhi, aged 80. Majhi has 15 words for river.

*Still spoken in Nepal, where Nepali is replacing it. [2]*

### Japan, traditional crafts

Registered craft workers: about 288,000 (1979) to 48,334 (FY2022).

*In Kanazawa, craft families close for lack of successors. [3]*

### Every company

42% of institutional knowledge is unique to the employee in their role.

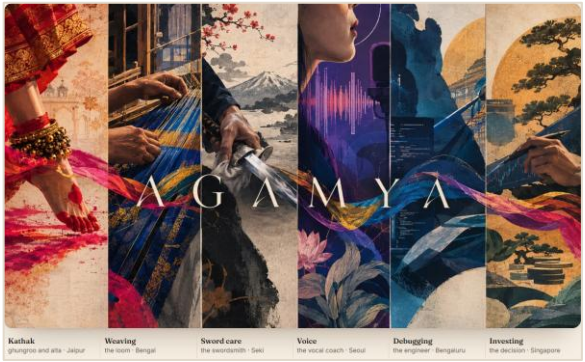
*Panopto survey, 2018. [4]*

**Agamya** captures the why, live, and passes it on

1 The expert works as usual, on screen or camera. **Moti**, a Gemini Live agent, asks one question at a pause.

2 A Work Map: steps, decisions, reasons in the expert's own verified words, and guardrails.

3 Learners practise with Moti as tutor, in their language. AI agents ask "would the expert stop here?"




[1] Daily Star 2026; Wikipedia [2] Down To Earth 2013; Wikipedia [3] Garland 2026, citing Assoc. for the Promotion of Traditional Craft Industries; NPR 2022 [4] Panopto 2018. Full references: slide 14 and notes.

# Opportunities

## How different is it from any of the other existing ideas?

| Others | Agamya |
|---|---|
| **Screen recorders, SOP writers**<br>record what was clicked | **ask why, live, at a pause** |
| **Document chatbots**<br>may invent answers | **the expert's verified quote, or "AI-inferred"** |
| **Training videos**<br>watch and forget | **practise; a tutor stops mistakes** |
| **Agent guardrails**<br>hand-written prompts | **from the expert, machine-checkable** |
| **English-first tools** | **Marathi, Hindi, Kannada and more** |

## How will it be able to solve the problem?

**1 Capture**
Moti asks why while the expert works

↓

**2 Understand**
Work Map with verified quotes

↓

**3 Confirm**
the expert approves and chooses who learns

↓

**4 Transfer**
practice, listening, Ask, AI agents

↓

**5 Measure**
impact from records, not estimates

*Minutes of the expert's time.*

## USP of the proposed solution

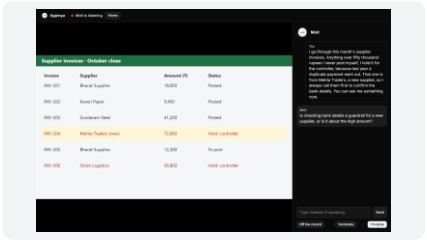
**One confirmed judgment, in the expert's own words, becomes three things:**

**a lesson**
for people, in 15 languages

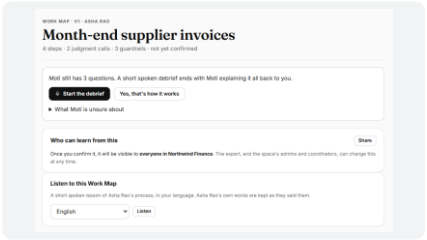
**a live tutor**
that stops a wrong decision

**a guardrail**
for AI agents (REST, MCP, ADK)

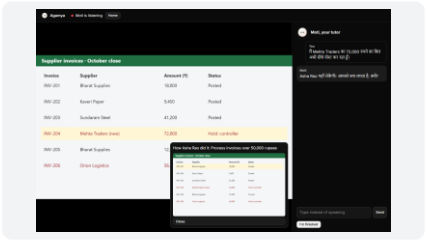
# List of features offered by the solution



**1. Live capture with Moti**
Gemini Live asks at a pause; off the record any time



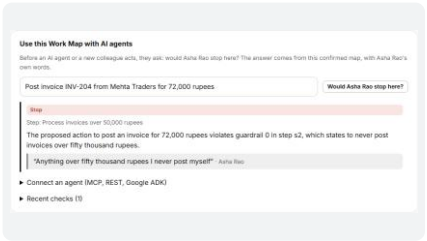

**2. Work Map**
steps, reasons in the expert's verified words, guardrails



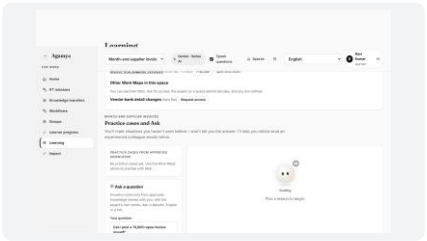
**3. Practise with a tutor**
camera or screen; Moti stops a wrong decision



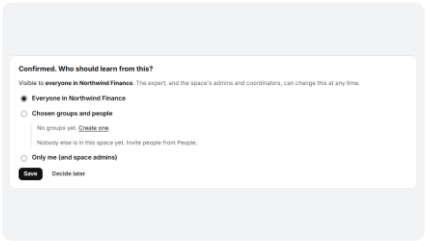

**4. Listen**
a spoken lesson in 15 languages



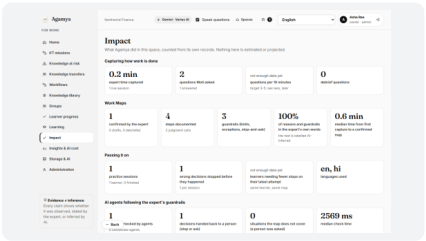

**5. AI Guardian**
STOP / ASK / PROCEED via REST, MCP, ADK



**6. Ask**
answers only from shared maps, quoting the expert




**7. Confirm and share**
space, groups or people; request access




**8. Impact, measured**
minutes, maps, mistakes stopped, AI cost

Also: debrief and teach-back · spaces and roles · optional space passcode · consent, masking of personal data, per-space isolation · works on phones (camera)

# Process flow diagram or Use-case diagram

**Expert works**
screen or camera

→

**Moti asks why**
one question at a pause

→

**Work Map**
verified quotes, guardrails

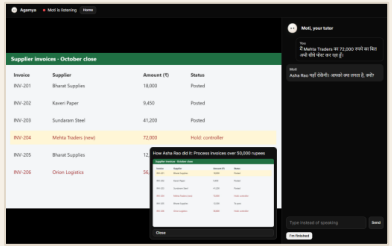
→

**Debrief**
Moti explains it back

→

**Expert confirms**
and chooses who learns

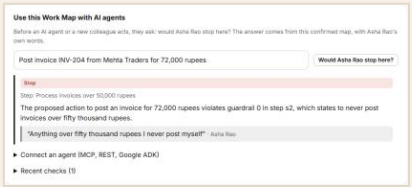
## Learners practise
Moti stops mistakes, in their language



## Listen and Ask
15 languages; answers quote the expert




## AI agents check
"Would the expert stop here?"




**Work:** accounts payable, IT runbooks, escalations, a retiring specialist

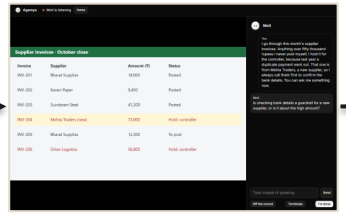
**Impact, measured**

**Culture:** a Kathak guru's footwork, a weaver's loom, a family recipe in Marathi

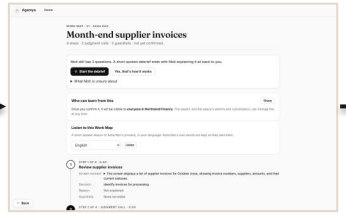
# Wireframes/Mock diagrams of the proposed solution (optional)




**1 Consent**
what is captured, who sees it



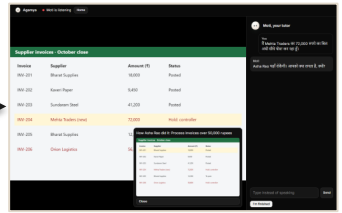
**2 Moti asks**
at a pause, by voice



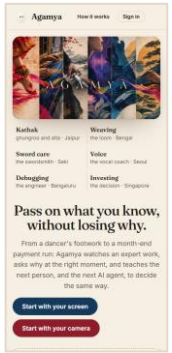

**3 Work Map**
steps, quotes, guardrails



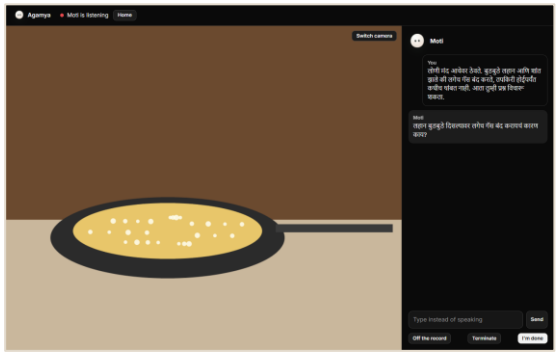

**4 Confirm, share**
who learns from this



**5 Practise**
Moti stops a mistake



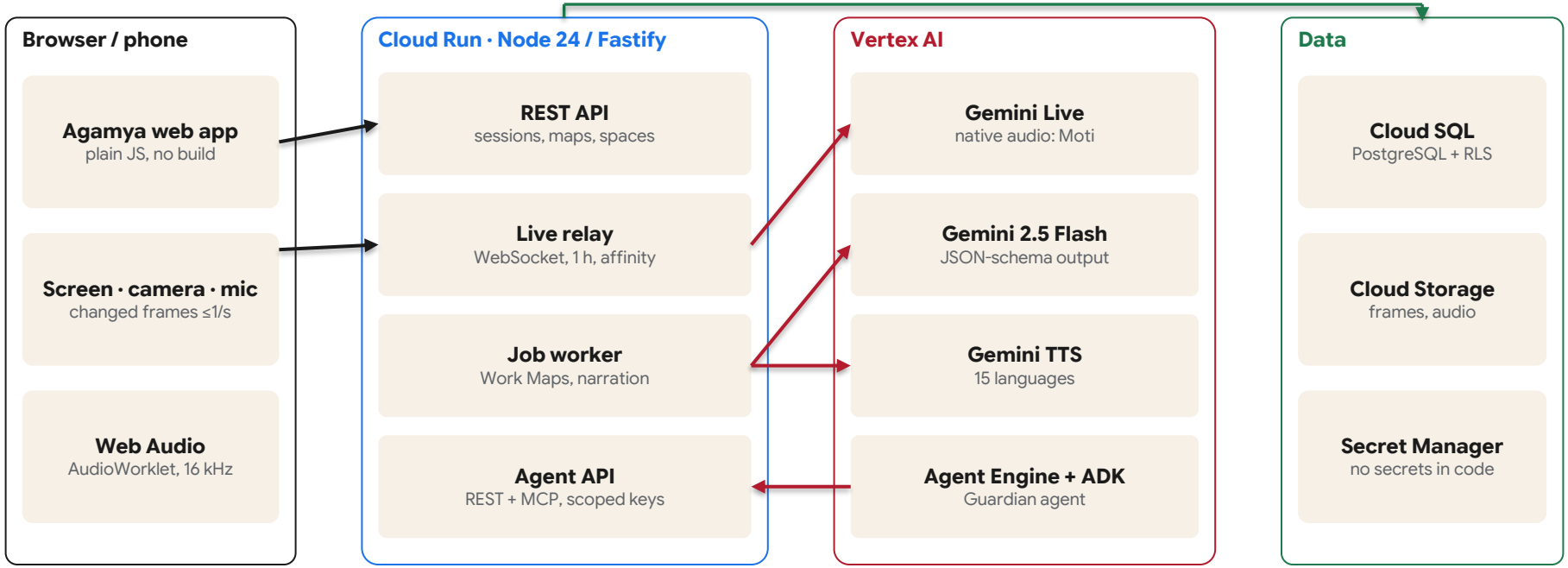

**On a phone**
Camera capture and practice; switch front and back camera. Tested on iPhone.




**Agamya for Culture**
The same loop on camera: here Moti asks in Marathi while a recipe is cooked.

*These are real screens of the deployed prototype, not mock-ups.*

# Architecture diagram of the proposed solution




**Flow:** audio and changed frames go through the relay to Gemini Live, and Moti's voice comes back. The worker builds Work Maps with Gemini 2.5 Flash; every quote is checked in code. External AI agents, and the ADK Guardian on Agent Engine, call the Agent API before acting. Row-Level Security isolates each space in the database.

# Technologies to be used in the solution

## Google Cloud, and why

| | |
|---|---|
| **Gemini Live API**<br>gemini-live-2.5-flash-native-audio | Moti: real voice conversation with vision, in Indian languages |
| **Gemini 2.5 Flash**<br>structured JSON output | Work Maps, tutor stops, agent checks, Ask: fast and cheap, checked in code |
| **Gemini TTS**<br>gemini-2.5-flash-tts | narration in 15 languages |
| **Vertex AI Agent Engine + ADK**<br>Python Guardian agent | managed agent hosting with tracing |
| **Cloud Run**<br>WebSockets, 1 h timeout, affinity | the whole app; scales to zero |
| **Cloud SQL · Storage · Secret Manager**<br>PostgreSQL 16 + RLS | isolation enforced by the database; private media; no secrets in code |
| **Cloud Build · Artifact Registry**<br>Cloud Logging | one-command deploy: npm run deploy |

## Also used

**Node.js 24 · Fastify**
server and WebSocket relay

**Plain JavaScript web app**
no build step, works on phones

**Web Audio · media capture**
AudioWorklet, screen and camera

**Model Context Protocol**
agents call the guardrails as a tool

**PGlite**
local development database

**axe-core · Playwright**
accessibility checks, screenshots

**Public datasets**
YouCook2, ai4privacy (evaluation)

# Estimated implementation cost (optional)

| | | |
|---|---|---|
| **Build one Work Map (Gemini 2.5 Flash)** | **≈ $0.005** | *Measured: 50 builds cost $0.23* |
| **10-minute live capture with Moti** | **≈ $0.10–0.20** | *Estimate at list prices* |
| **Spoken lesson, per language (cached)** | **≈ $0.01** | *Estimate at list prices* |
| **One agent check** | **< $0.001** | *Estimate at list prices* |
| **Cloud Run** | **pay per use** | *scales to zero* |
| **Cloud SQL (db-f1-micro) + Storage** | **≈ $10–15/mo** | *pilot size, estimate* |

**Example**

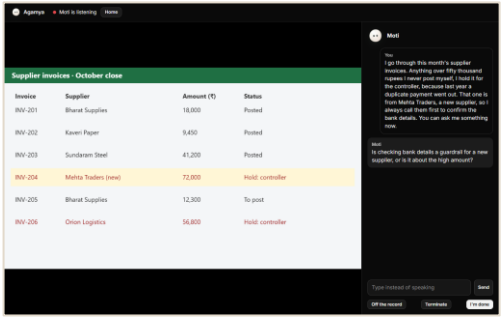
One expert, 20 workflows, about 30 minutes each:

## under $15 of AI

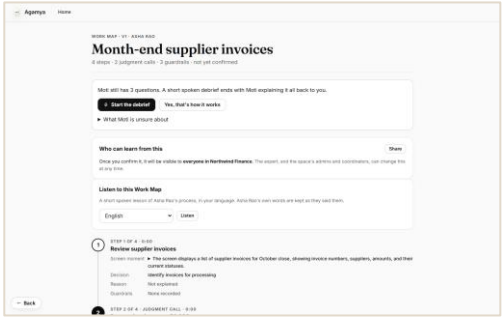
for a fully mapped role (estimate)

**How we know:** every AI call is logged with its list-price cost, and the Impact page shows AI cost per map. Green = measured; gold = estimate.

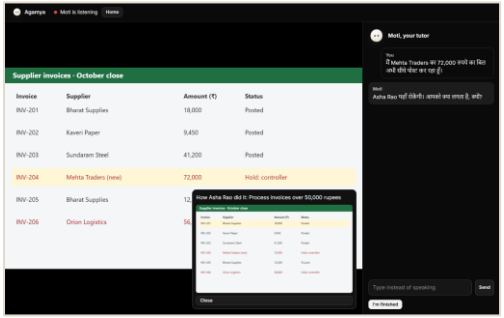
# Snapshots of the prototype



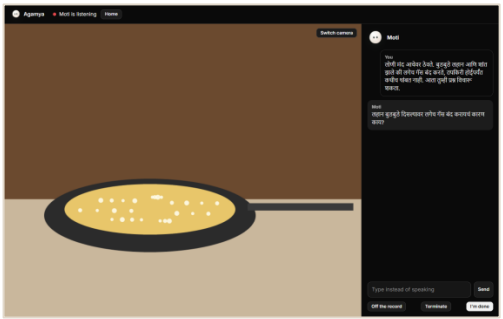

Moti asks a guardrail question



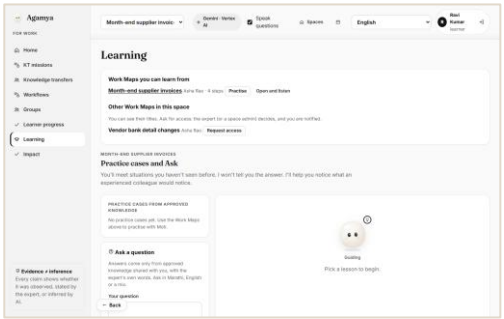

Work Map, verified quotes



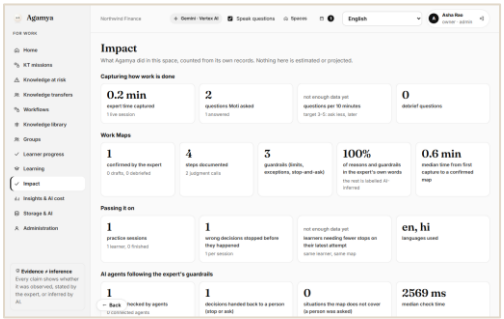

Tutor stops a mistake (Hindi)




Culture: Moti in Marathi




Learning and Request access




Impact, measured

# Prototype Performance report/Benchmarking

| 176 / 176 | 22 / 22 | ≈ 2 s | ≈ 1.6 s |
|---|---|---|---|
| automated tests pass | end-to-end checks on the live app | Moti asks after the expert pauses | tutor stops a wrong decision |

## Work Maps on 25 real recipes (YouCook2, 212 steps)

| 100% | 0% | 100% |
|---|---|---|
| steps covered | invented evidence | steps in order |

**Quotes the expert didn't say:** silent videos 1 claimed → 0 verified; narrated 3 → 3. Unspoken quotes are labelled AI-inferred, every time.

**Cost:** 50 Gemini calls, $0.23.

**Guardian agent on Agent Engine:** returned STOP for ₹88,000 and ₹64,500 postings over the expert's limit (live test).

## Personal data masked before storage




ai4privacy, 43,501 sentences. Amounts and dates: 0% changed.

All figures measured by the team (see docs/EVALUATION.md). Latencies are from a small number of live sessions. Also tested on iPhone.

# Additional Details/Future Development (if any)

| Next 3 months | 6 months | 12 months |
|---|---|---|
| Two experts on one map: where they agree, differ, and why | Export guardrails to agent platforms and RPA tools | Culture archive: opt-in community collections of crafts and languages |
| Native-speaker review of all 15 narration languages | On-device masking of faces and screens before upload | Partnerships with craft associations and language surveys |
| Google Workspace sign-in and directory groups | Expert-shift alerts when a confirmed rule changes | Certified practice paths for regulated roles |




## Our Inspiration

The app has a story page: Bengal muslin, Majhi, Japan's craft families, and the expert at your desk, with sources. The same loop that keeps an invoice rule can keep a weaver's reason.

agamya-82538920788.us-central1.run.app/#/inspiration

# Provide links to your:

**GitHub Public Repository**
github.com/S-D-P/Agamya-AI-Builder-Cup-26

**Final Product Link**
agamya-82538920788.us-central1.run.app

**Demo Video Link (4 Minutes)**
Video Link

**Culture Mode Demo Video**
Video Link

**Work Mode Demo**
Video Link

**Learner Demo**
Video Link

**Judge sign-in**

One click fills these in on the sign-in page.

**Work:** judge.work@agamya.example
**Culture:** judge.culture@agamya.example
**Password:** agamya-judges-2026

**Start here:** docs/JUDGES_GUIDE.md in the repository shows every feature, one screenshot at a time.

Use Chrome or Edge on a computer for screen sharing; on a phone, use the camera. Allow the microphone and wear headphones.

# References

[1] The Daily Star (2026). What Bengal made and let others claim. thedailystar.net · Wikipedia: Dhaka muslin.

[2] Chakravartty, A. (6 Sep 2013). Survey tracks down only speaker of Majhi in India. Down To Earth · Wikipedia: Majhi language.

[3] Garland Magazine (2026). Kimono and lacquerware sales down up to 90 per cent as Japan's craft base narrows (data: Association for the Promotion of Traditional Craft Industries) · NPR/WFAE (29 Jul 2022). Japan's traditional crafts are struggling to survive the country's population decline.

[4] Panopto (2018). Workplace Knowledge and Productivity Report, via HR Dive.

[5] Polanyi, M. (1966). The Tacit Dimension.

[6] Zhou, L., Xu, C., Corso, J. (2018). YouCook2 dataset · ai4privacy. pii-masking-200k (Hugging Face).

Agamya's own results: docs/EVALUATION.md. Full URLs in these notes and in docs/submission/RESEARCH_LEDGER.md.

Google Cloud H2S

# AI Builder Cup

Build in JAPAC.
Build for the world.

# Thank you

